

5(a)	any five from: 1 evolution / speciation / natural selection ; 2 random / spontaneous ; 3 mutations (occurred) ; 4 adapting some plants to, alpine habitat / high altitude / mountains ; 5 (in this habitat) these plants, survived / were selected for ; 6 (selection pressure is / must resist) cold / strong winds / thin soil ; 7 alleles for alpine features passed to offspring ; 8 change in, allele frequency / gene pool (of population) ; 9 population become, morphologically / physiologically / biochemically, different ; 10 less, herbivores / competition, in new <u>niche</u> ; 11 AVP ;	5
5(b)(i)	any three from: 1 time period = from 0.95 million years (to present) ; 2 mountains high enough to exceed treeline ; 3 this provides alpine, habitat / region / niches ;	3

Question	Answer	Marks
5(b)(ii)	any two from: 1 compare / find similarities in / find differences in, DNA ; 2 of alpine species and related (lowland) species ; 3 find (time) when species, evolved / separated / diverged / speciated ; 4 many, DNA / base / nucleotide, differences / changes / mutations, means they, separated a long time ago / have a distant common ancestor ; 5 AVP ;	2